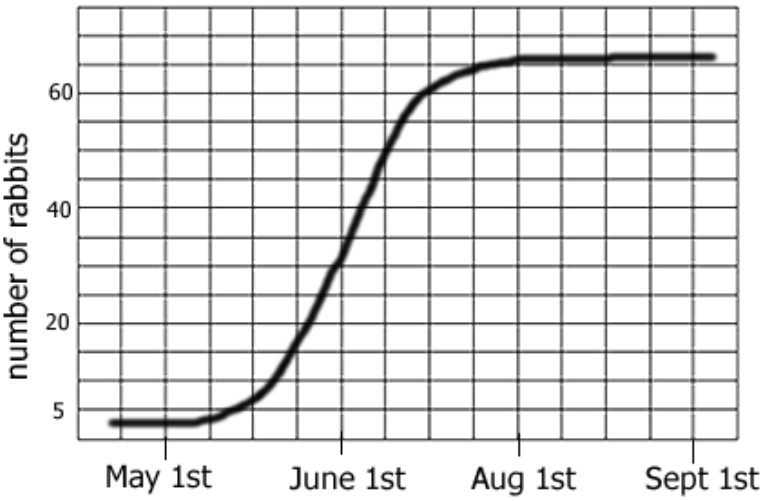
http://www.biologycorner.com/worksheets/interpreting_data.html

Name:_____ Date:_____

INTERPRETING ECOLOGICAL DATA

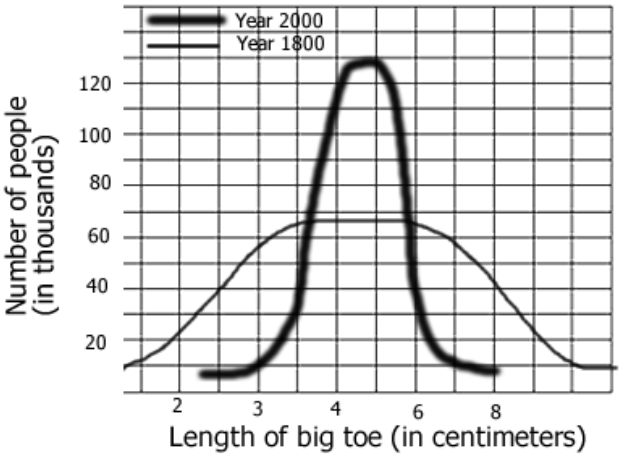
Graph 1: Rabbits Over Time



a. The graph shows a _____ growth curve.

b. The carrying capacity for rabbits is _____

c. During which month were the rabbits in exponential growth?



Graph 2: Average Toe Length

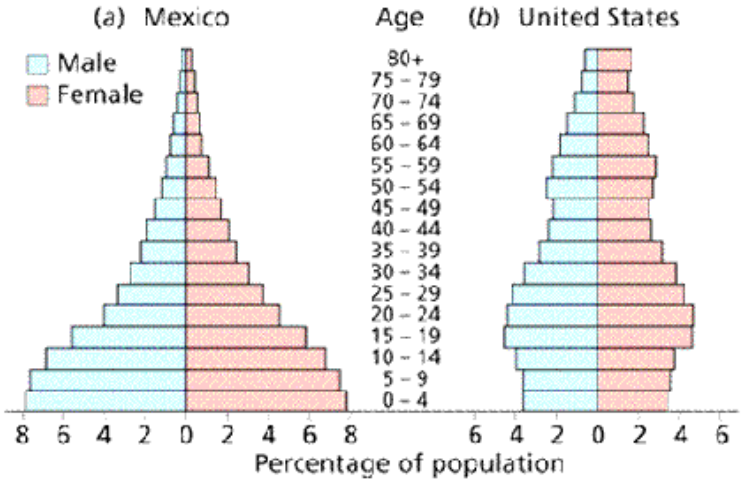
a. In 1800, about how many people surveyed had a 3 cm toe? _____

b. How many in 2000? _____

c. In 2000, what is the average toe length? _____

What is the average toe length in 1800 _____

Graph 3: Mexico and US



a. In Mexico, what percentage of the population is between 0-4 years of age? _____ In the US? _____

b. Which population is growing the fastest? _____

c. Which age group has the smallest number in both countries? _____

Chart 4: Trapping Geese

In order to estimate the population of geese in Northern Wisconsin, ecologists marked 10 geese and then released them back into the population. Over a 6 year period, geese were trapped and their numbers recorded.

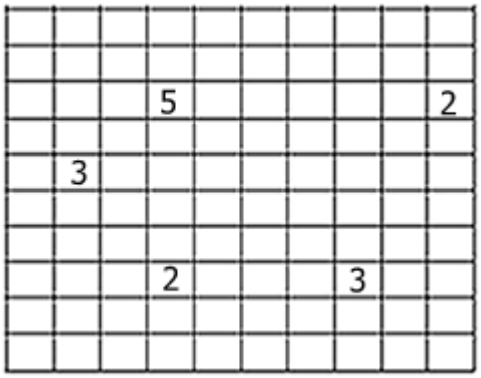
a. Use the formula to calculate the estimated number of geese in the area studied? _____

b. Supposing more of the geese found in the trap had the mark, would the estimated number of geese in the area be greater or lesser? _____

Year	Geese Trapped	Number with Mark
1980	10	1
1981	15	1
1982	12	1
1983	8	0
1984	5	2
1985	10	1

$$\frac{\text{(Total number captured) x (number marked)}}{\text{(total number recaptured with mark)}}$$

Chart 5: Mushroom Plots



Another ecologist uses a different method to estimate the number of mushrooms in a forest. She plots a 10x10 area and randomly chooses 5 spots, where she counts the number of mushrooms in the plots and records them on the grid.

a. Calculate the number of mushrooms in the forest based on the grid data – Show all your working

Chart 6: Snakes & Mice

The data shows populations of snake and mice found in an experimental field.

Year	Snakes	Mice born	Mice died	Mice
1960	2	1000	200	600
1970	10	800	300	1400
1980	30	400	600	1300
1990	15	600	550	1350
2000	14	620	600	1370
2001	15	640	580	1430

a. During which year was the mouse population at zero population growth? _____

b. What is the approximate carrying capacity for snakes? _____

c. What is the carrying capacity for mice? _____

d. What is the rate of growth (r) for mice during 1970? _____ During 1980? _____

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